

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

NEWTON'S LAWS OF MOTION

LECTURE-5

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

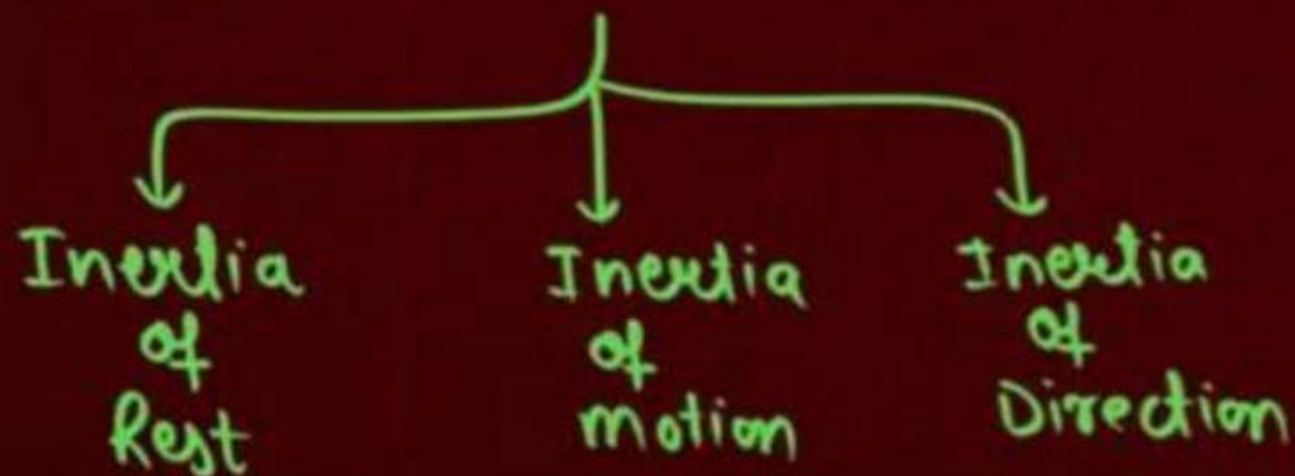
- 1** Revision ✓
- 2** Gabbar Numerical Series ✓
- 3** ..
- 4** ..

HARD WORK BEATS TALENT ✓
WHEN TALENT DOESN'T ✓
WORK HARD

Recap From 1st Lecture

Inertia (1st law of motion)

Property of Body by virtue of which Body cannot change its state of **Rest**, **motion** and **Direction**



Momentum (p)

$$\vec{p} = m\vec{v} \quad \text{kg m/s Vector}$$

$$\bullet p_f = mv \quad \bullet p_i = mv$$

② # Newton 2nd laws of Motion

$$F = \frac{\Delta p}{\Delta t} = ma \rightarrow \text{qth}$$

$$F = \frac{\Delta p}{\Delta t} = \frac{dp}{dt} = m \frac{dv}{dt} + v \frac{dm}{dt}$$

③ **Impulse**

Force acting for the short period of time.

$$\uparrow F = \frac{\Delta p}{\Delta t} \downarrow$$

$$F = \frac{dp}{dt} \quad \{ F dt \} = dp$$

Impulse

Impulse momentum theorem

Recap From 2nd Lecture

Newton third law of motion

Action = -Reaction

- * Rocket propulsion
- * walking
- * Boat
- * Recoiling on gun
- * Flying of Bird.

law of conservation of momentum

$$\text{if } F_{\text{ext}} = 0$$

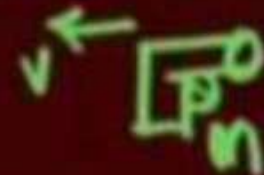
$$P_i = P_f$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

Application of law of conservation of momentum

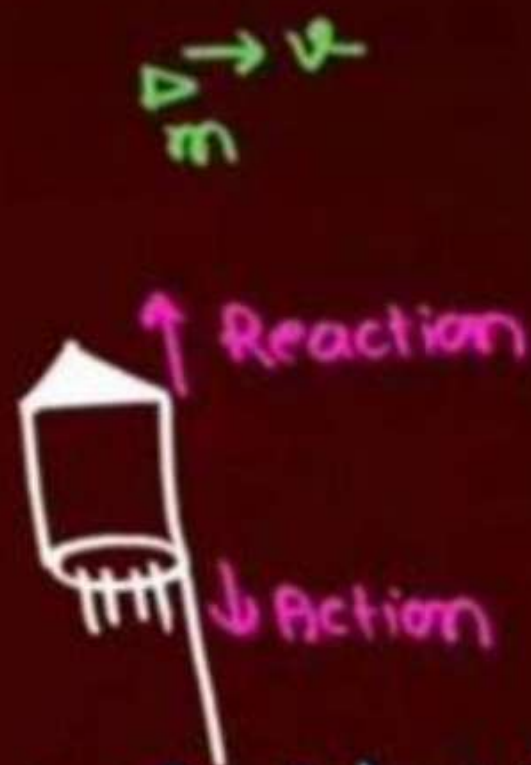
① Recoiling velocity of gun

$$V = -\frac{mv}{M}$$



② Rocket propulsion

$$F = m \frac{dv}{dt} + v \frac{dm}{dt}$$



? why Newton 2nd law is also Real law of Motion

$$F = \frac{dP}{dt}$$

2nd law derive $V = u$

3rd law Derived
 $f_{12} = -f_{21}$

Lecture no. 3

Types of forces.

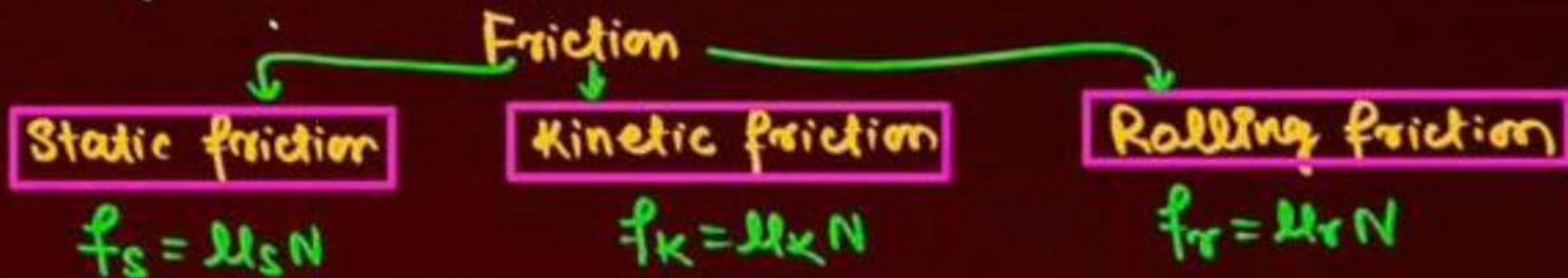
- * Gravitation force
- * Spring force
- * Tension
- * Normal Force
- * Electrostatic force
- * Friction force ✓

Friction

- * Force oppose the Relative Motion $f = \mu N$
- * Roughness & Adhesion

Laws of friction

- $f = \mu N$
- friction is independent of Area.

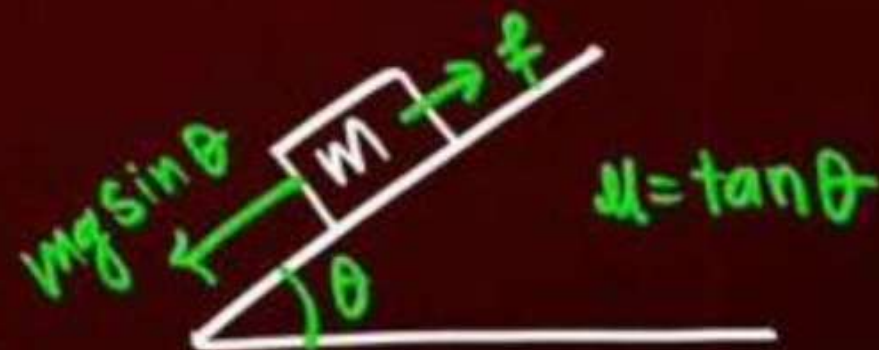


- ✓ Advantage of friction ✓
- ✓ Disadvantage of friction ✓
- ✓ How to Reduce the friction. ✓

Angle of friction

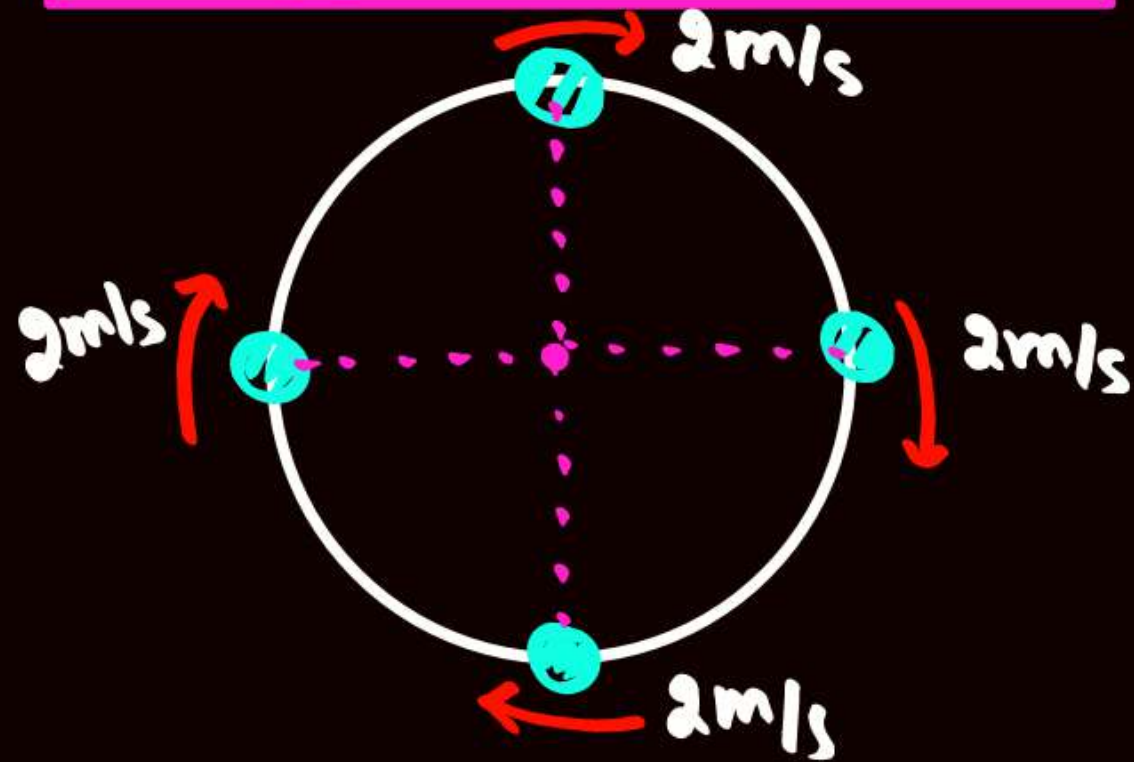


Angle of Repose



(lecture. no. 4)

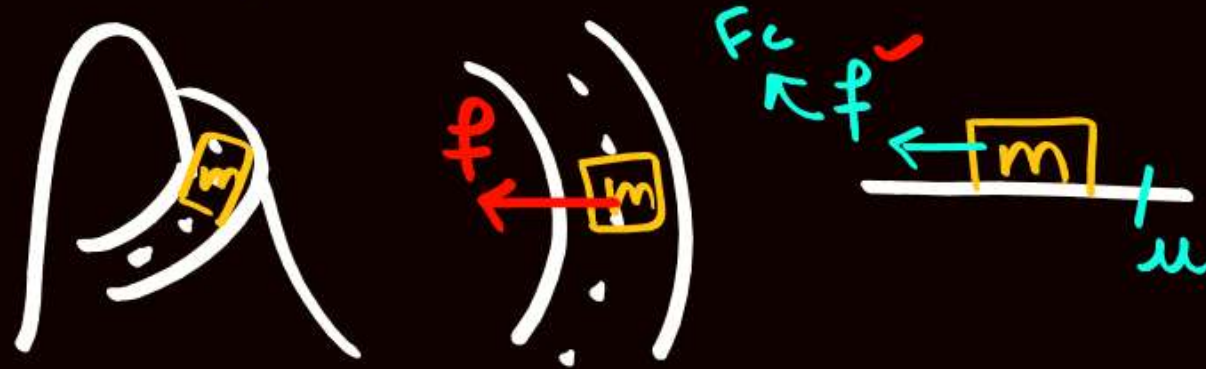
Centripetal acc/r



- # Speed constant
- # velocity change

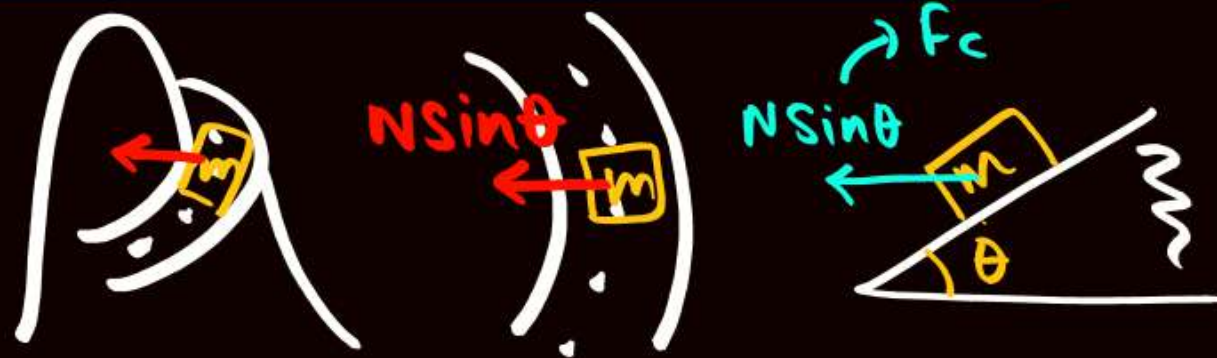
$$a_c = \frac{v^2}{r} \quad F_c = ma_c = m \frac{v^2}{r}$$

① Turning on friction Road



$$v = \sqrt{rg\mu}$$

② Turning on Banked Road



$$v = \sqrt{rg \tan \theta}$$

③ Turning on friction Road and Banked Road



$$v = \sqrt{rg \left(\frac{\mu + \tan \theta}{1 - \mu \tan \theta} \right)}$$

Puri Taan

Gabbar Numerical Series PYQs

QUESTION [SUMMER SERIES-A 2026]

An impulse is the change in:

- A** Velocity
- B** Kinetic energy
- C** Momentum ✓
- D** Displacement

QUESTION [SUMMER SERIES-A 2026]

3mark *derive*

✓ State and explain Newton's Second Law of Motionl. Hence deduce relation $F = ma$.

QUESTION [SUMMER SERIES-B 2026]

mca m=1



If no external force acts on a system of particles, its total linear momentum will-

$$\Delta P = 0$$

- A** Be zero
- B** Increase gradually
- C** Decrease gradually
- D** Remain constant ✓

$$F = \frac{\Delta P}{\Delta t}$$

$$0 = \frac{\Delta P}{\Delta t}$$

$$0 \times \Delta t = \Delta P$$

$$0 = \Delta p$$

$$0 = p_f - p_i$$

$$p_i = p_f$$

$$p_i = p_f$$

$$2 = 2$$

no momentum is not
changed

constant

QUESTION [SUMMER SERIES-B 2026]

Q- State and explain Newton's third Law of Motion. ✓ 3 marks

QUESTION [SUMMER SERIES-B 2026]

3 mark

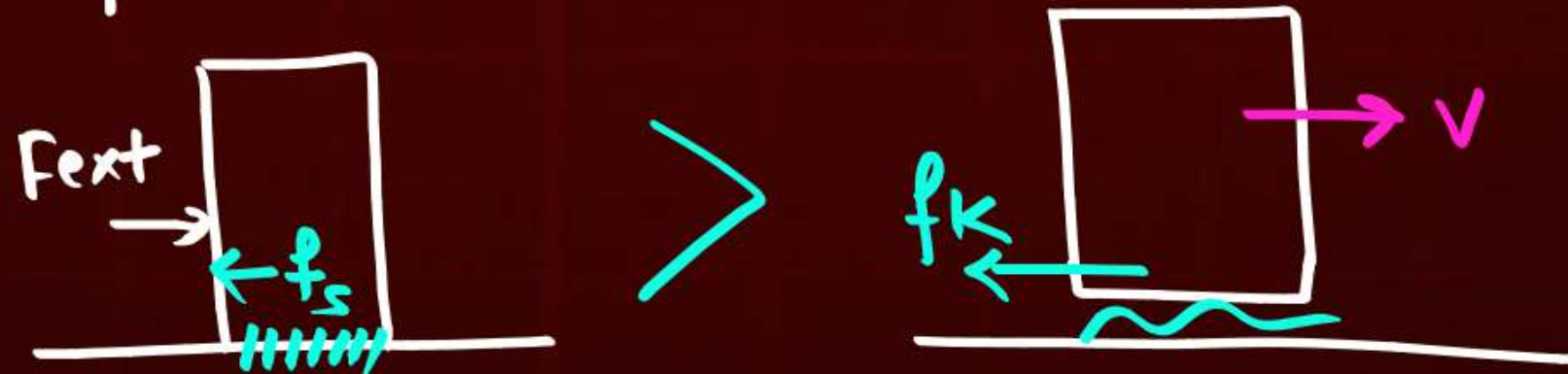
Explain static friction and kinetic friction. Can the value of kinetic friction be greater than static friction? Explain.

✓
✓
✓
✗ $f_k > f_s$

$f_s > f_k$

NO, the value of Static friction is always greater than the value of Kinetic friction

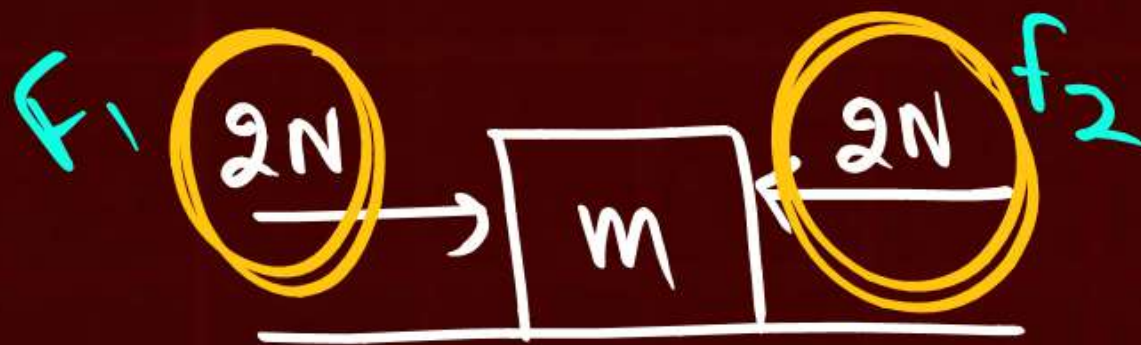
Because when the Body moves the contact b/w the surface decrease hence the Kinetic friction decrease.



QUESTION [SUMMER SERIES-B 2026]

Briefly explain equilibrium of a rigid body.

Ans The Body is said to be in equilibrium
when $F_{\text{net}} = 0$



$$\begin{aligned}
 F_{\text{net}} &= F_1 + F_2 \\
 &= 2 + (-2) \\
 &= 2 - 2 \\
 &= 0 \checkmark
 \end{aligned}$$

QUESTION [SUMMER SERIES-C 2026]

A rocket's motion is based on the principle of:

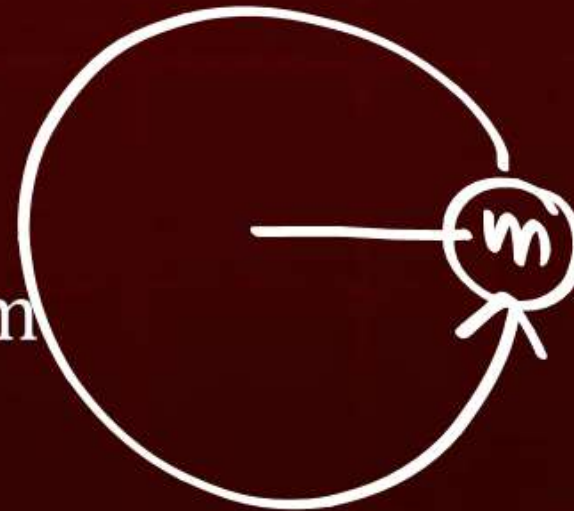
- A** Gravitational force
- B** Conservation of kinetic energy
- C** ✓ Conservation of linear momentum
- D** Conservation of angular momentum

✓ # Linear momentum / momentum P

$P = mv$



~~#~~ Angular momentum $\vec{L}$



QUESTION [SUMMER SERIES-C 2026]

✓ What are three types of inertia? Give one example of each.

QUESTION [WINTER SERIES-A 2025]

Momentum is measured in:

(P)

unit

$$p = \underline{m} \underline{v}$$

$$= \text{kg m/s}$$

$$= \text{kg m s}^{-1}$$

A Kgs m s⁻¹

B Kgm s⁻²

C Kgm s⁻¹

D Kgm

QUESTION [WINTER SERIES-A 2025]

State the physical significance of (moment of inertia.)

→ Rotation

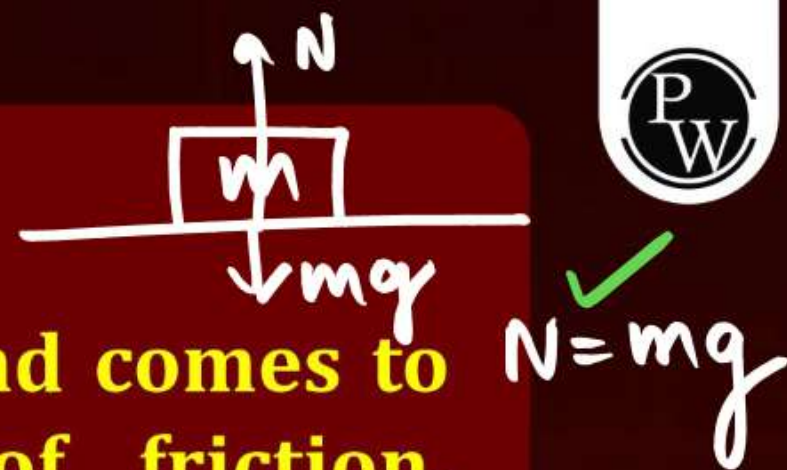
$$I = mr^2$$

Inertia

moment of Inertia

QUESTION [WINTER SERIES-A 2025]

✓ A cricket ball is rolled on ice with a velocity of 5.6 m/s and comes to rest after travelling 8 metre. Find the coefficient of friction. ($g = 9.8 \text{ m/s}^2$)



Sol

$$u = 5.6 \text{ m/s}$$

$$\mu = ?$$

$$0 = 31.36 + 16a$$

$$-31.36 = 16a$$

$$\frac{-31.36}{16} = a$$

$$-1.96 \text{ m/s}^2 = a$$

Retardation

$$F = f$$

$$ma = \mu N$$

$$ma = \mu mg$$

$$1.96 = \mu \times 9.8$$

$$\frac{1.96}{9.8} = \mu$$

$$0.2 = \mu$$

$$a = ?$$

$$v^2 = u^2 + 2as$$

$$0^2 = (5.6)^2 + 2a8$$

Surface
ice

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

QUESTION [WINTER SERIES-A 2025]

State Newton's first and second laws of motion, also define one Newton of force.

Sol: 1 Newton force

$$F = \underline{m}a$$

$$= 1\text{kg} \times 1\text{m/s}^2$$

$$= 1\text{kg m/s}^2$$

$$= \underline{\underline{1\text{N}}}$$

 A 1 Newton force is the force which produce the acc of 1m/s² in 1kg body."

QUESTION [WINTER SERIES-C 2025]

A rocket works on the principle of conservation of: (Repeat)

- A** Linear momentum ✓
- B** Angular momentum ✗
- C** Charge ✗
- D** Force ✗

QUESTION [SUMMER SERIES-A 2025]

Motion 1D



When the distance travelled by a body is proportional to the time taken, what happens to its speed? (Neet-2017)

$$\frac{dt^1}{dt} = 1 t^{1-1} = 1 \times t^0 = 1 \times 1 = 1$$

Sol $x \propto t$
 $x = ct$

$$\text{Speed} = \frac{dx}{dt} = \frac{dct}{dt} = c \frac{dt}{dt} = c \times 1 = c$$

= Constant
= do not change

A

Becomes zero

B

Increases

C

Remains the same

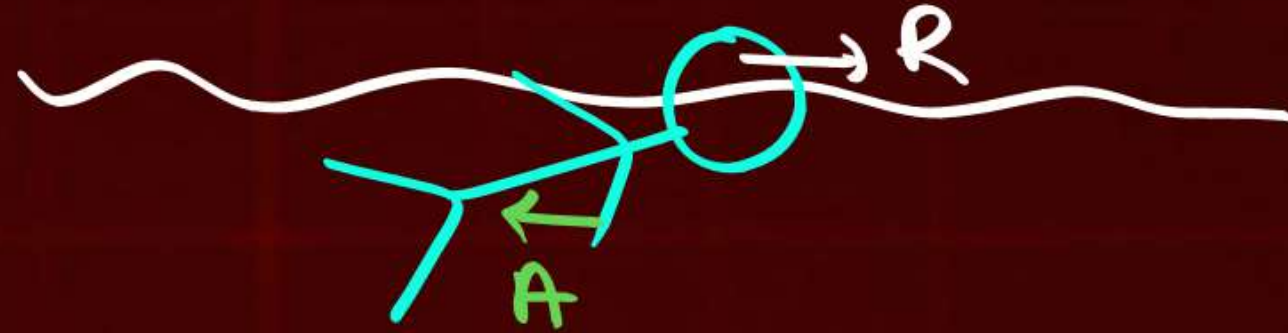
D

Decreases

QUESTION [SUMMER SERIES-A 2025]

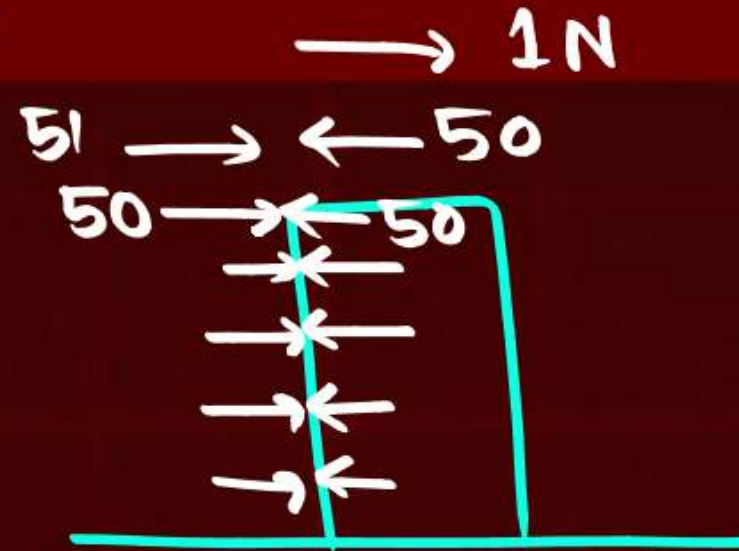
The forward movement in swimming takes place because of:

- A** First law of motion
- B** Second law of motion
- C** Third law of motion ✓
- D** Fourth law of motion



QUESTION [SUMMER SERIES-A 2025]

The maximum value of static friction when the body is at the verge of starting motion is known as:



- A** Impending motion
- B** Angle of repose
- C** Static friction
- D** Limiting friction ✓

QUESTION [SUMMER SERIES-A 2025]



State the conditions for the equilibrium of a particle under concurrent forces.

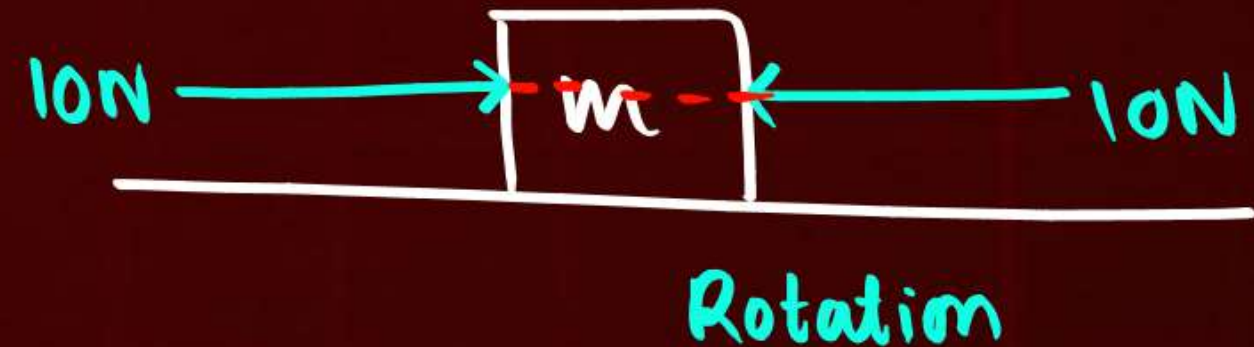
doubt

When two equal and opposite force act on same Body



$F_{ext} = 0$

The force which have same line of Action



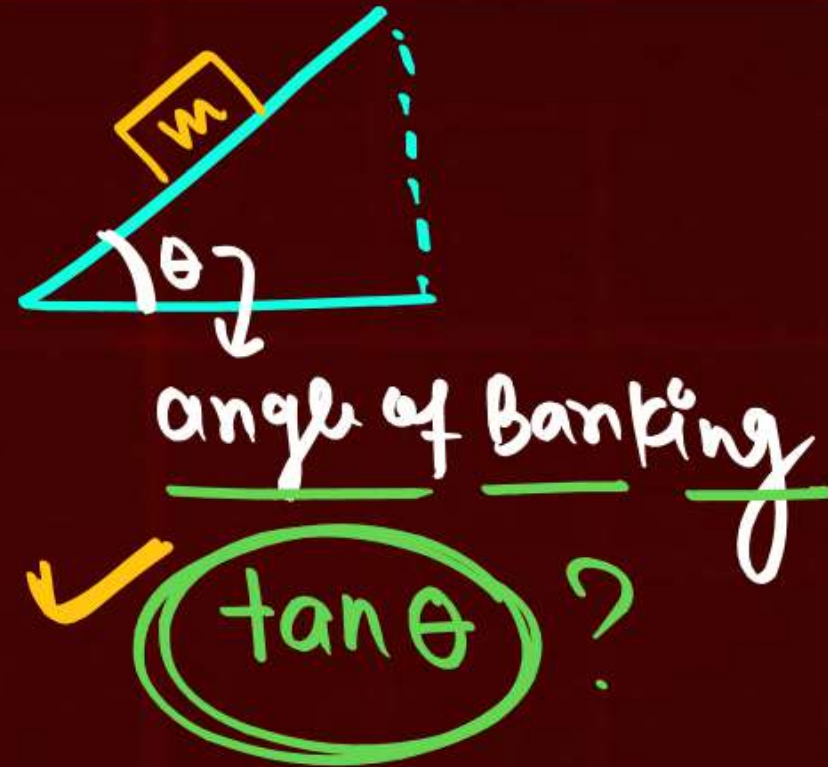
QUESTION [SUMMER SERIES-A 2025]

✓ Why are the curved roads banked? Find the angle of banking of curved road.

Ans



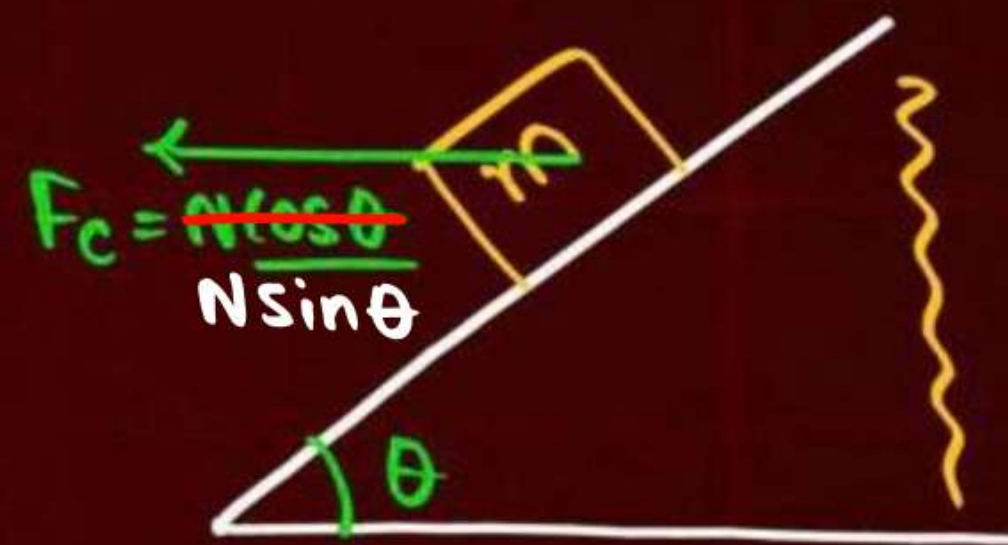
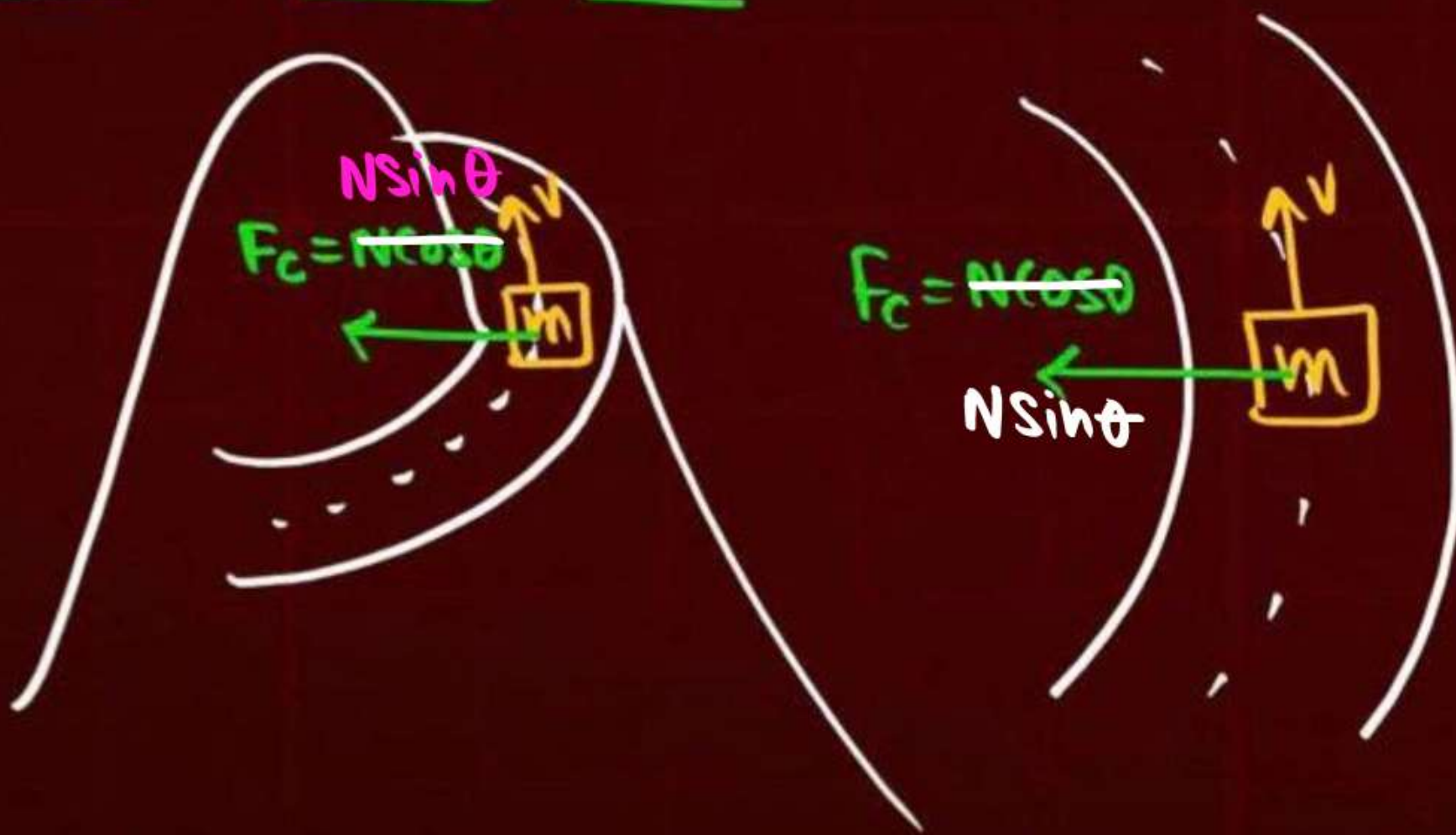
- # To avoid the accident
- # The vehicle/car remain in circular path.

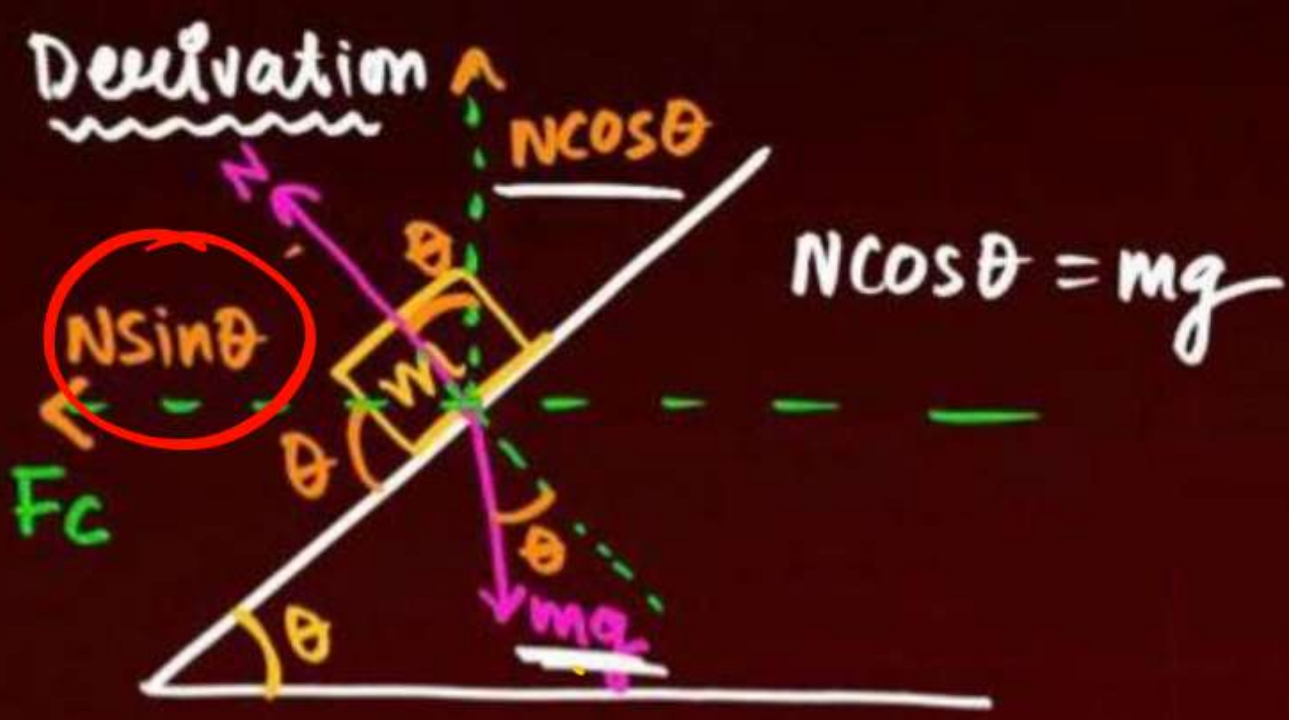


③ Maximum safe velocity on banked road

(Turning) No friction.

When a road is banked (tilted at an angle), the vehicle can take a turn more safely because a part of the normal force provides the required centripetal force-so it doesn't rely only on friction.





$N \sin \theta = F_c$
 $N \sin \theta = \frac{mv^2}{r}$ — (I)
 $N \cos \theta = mg$ — (II)

Divide (I) and (II)

$$\frac{N \sin \theta}{N \cos \theta} = \frac{\frac{mv^2}{r}}{mg}$$

$$\tan \theta = \frac{mv^2}{mg r}$$

$$\tan \theta = \frac{v^2}{rg}$$

$$rg \tan \theta = v^2$$

$$\sqrt{rg \tan \theta} = v$$

Numerical

friction

$$v = \sqrt{rg \mu + rg \tan \theta}$$

Banked

✓ Angle of Banking

$$\checkmark \tan \theta = \frac{v^2}{rg}$$

$$\theta = \frac{v^2}{\tan^{-1} rg}$$

$$\checkmark \theta = \tan^{-1} \left(\frac{v^2}{rg} \right)$$

QUESTION [SUMMER SERIES-B 2025]



$$F=ma$$

Which law of newton helps in finding the force on a boyd?

- A** First law
- B** Second law ✓
- C** Third law
- D** None of these

QUESTION [SUMMER SERIES-B 2025]

Limiting friction of a body depends on:

friction $\rightarrow N$ ✓
 $\rightarrow$ nature of surface

$\rightarrow$ do not depend on Area & volume

- A** Area of contact of surface
- B** Volume of smaller body on larger surface
- C** The periphery of the contact surface
- D** Nature of surfaces

QUESTION [SUMMER SERIES-B 2025]

Deduce the relation $F = \underline{ma}$ from second law of motion. ✓

QUESTION [SUMMER SERIES-C 2025]

Which one of the following is not an example of the third law of motion?

- A** Walking ✓
- B** Cycling ✓
- C** Skiing ✓
- D** Walking on a boat ✓



QUESTION [SUMMER SERIES-C 2025]

Which of the following is self adjusting force?

- A** Kinetic friction
- B** Limiting friction
- C** Static friction ✓
- D** All of these

QUESTION [SUMMER SERIES-C 2025]

Explain term impulse. Give one example.

QUESTION [SUMMER SERIES-X 2024]

State Newton's three laws of motion.

3 laws ✓

3 mark

QUESTION [SUMMER SERIES-Y 2024]

lecture no 2

Show that Newton's second law is the real law of motion.

QUESTION [SUMMER SERIES-Z 2024]

Define Friction. State laws of limiting friction. ✓

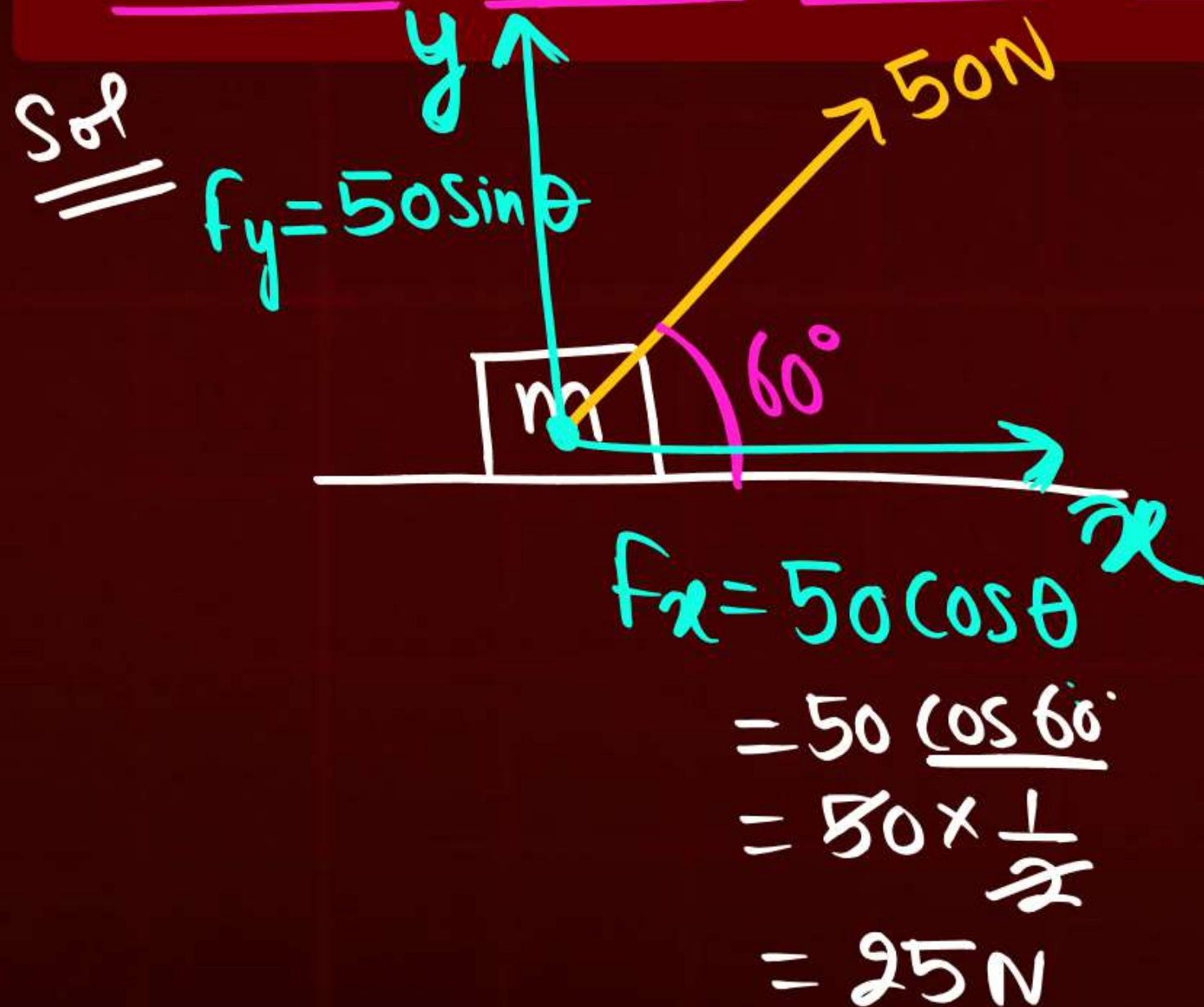
$$f \rightarrow N$$

$$f \rightarrow \mu$$

$$f \rightarrow A^x$$

QUESTION [SUMMER SERIES-Z 2024]

- ✓ A force of 50 N acts on a particle at an angle of 60° with horizontal. Calculate its two rectangular components.



$$\cos 60 = \frac{1}{2}$$

$$\sin 60 = \frac{\sqrt{3}}{2}$$

$$f_y = 50 \sin 60$$

$$= 50 \times \frac{\sqrt{3}}{2}$$

$$= 25\sqrt{3}$$

QUESTION [JEE MAIN 2023]

$$\checkmark \quad \underline{1i} + \underline{xj}$$



At any instant the velocity of a particle of mass 500 g is $(2t\hat{i} + 3t^2\hat{j})\text{ms}^{-1}$. If the force acting on the particle at $t = 1\text{s}$ is $(\hat{i} + x\hat{j})\text{N}$. Find x .

$$\begin{aligned} \text{So, } m &= 500 \text{ gm} \\ m &= \frac{500}{1000} \text{ kg} \\ &= \frac{5}{10} = \frac{1}{2} \text{ kg} \end{aligned}$$

$$\frac{dx^n}{dx} = \underline{nx^{n-1}}$$

$$\underline{v = 2t\hat{i} + 3t^2\hat{j}}$$

$$a = \frac{dv}{dt}$$

$$a = \frac{d(2t\hat{i} + 3t^2\hat{j})}{dt}$$

$$a = \frac{d}{dt} 2t\hat{i} + \frac{d}{dt} 3t^2\hat{j}$$

$$a = 1 \times 2t^{1-1}\hat{i} + 2 \times 3t^{2-1}\hat{j}$$

$$a = 2t^0\hat{i} + 6t^1\hat{j}$$

$$a = 2 \times 1\hat{i} + 6t\hat{j}$$

$$a = 2\hat{i} + 6t\hat{j} \quad (t = 1\text{ sec})$$

$$a = 2\hat{i} + 6 \times 1\hat{j}$$

$$a = 2\hat{i} + 6\hat{j}$$

$$F = ma$$

$$f = \frac{1}{2} (2\hat{i} + 6\hat{j})$$

$$F = \frac{1}{2} \times 2\hat{i} + \frac{1}{2} \times 6\hat{j}$$

$$F = 1\hat{i} + 3\hat{j}$$

$$F = \hat{i} + x\hat{j}$$

$$F = 1\hat{i} + x\hat{j}$$

also

$$\rightarrow F = 1\hat{i} + \underline{3}\hat{j}$$

$$\boxed{x=3}$$

QUESTION [NEET 2017]



$$g = 10 \text{ m/s}^2$$

A car is moving on a flat curved road of radius $r = 50 \text{ m}$.

The coefficient of friction between the tires and road is $\mu = 0.2$.

Find the maximum safe velocity.

$$v = \sqrt{rg\mu}$$

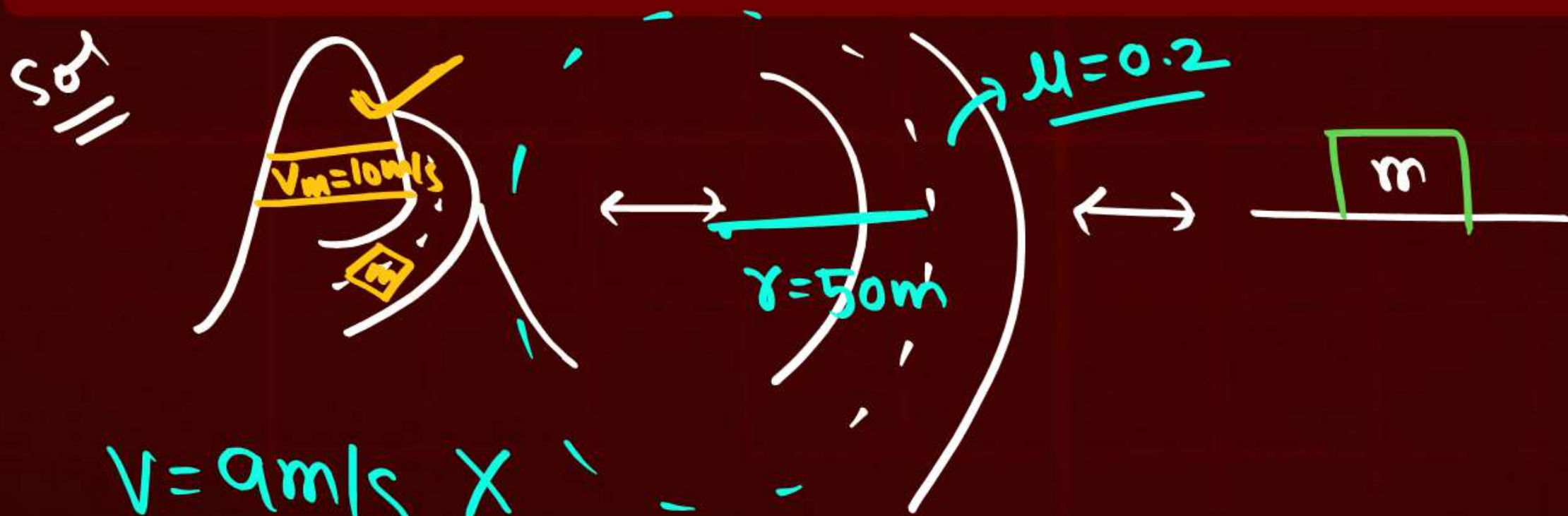
$$= \sqrt{50 \times 10 \times 0.2}$$

$$= \sqrt{50 \times 2}$$

$$= \sqrt{100}$$

$$= \sqrt{10^2}$$

$$= 10 \text{ m/s}$$



$$v = 9 \text{ m/s} \times$$

$v = 11 \text{ m/s} \checkmark$ Yes accident

JADUGAR QUESTION

1. State and explain Newton's 2nd law of motion why it is called real law of motion and what is impulse? ✓
2. What is friction. Explain why it is a necessary evil. What is its cause. Give its direction and the factors on which it depends. ✓
3. Define static, kinetic and rolling friction. ✓
4. Define centripetal force. Derive its expression. ✓
5. What is banking of road. Why is it necessary. Derive an equation for safe velocity on banked and frictional road

$$\frac{f + B}{\text{✓}}$$

SUMMARY

Numericals
PYQ



HOMework

CP

Copy → notes



Chapter completed~

Thank
You